

Electrical AC BOM (Phase 1)

As-of date: 2026-08-27

Purpose: maintain the installed Phase 1 AC architecture baseline: portable 30A EMS, 30A shore inlet/locking cord with modular 25 ft TT-30 extension, one 6-way DIN enclosure, 30A AC-in breaker, 30A AC-out main breaker, and two active 20A GFCI-protected AC-out branches. Owner report 2026-08-27 : each branch has one origin GFCI plus one downstream duplex, four duplex devices/eight plug positions total. Final acceptance still requires device/box/cover/clamp/fill inspection, LINE/LOAD/PE proof, downstream protection, GFCI trip/reset in inverter and accepted-shore modes, representative loading, and service-face/dead-front closeout.

Related docs:

- docs/implementation/ELECTRICAL_overview_diagram.md
- docs/core/SYSTEMS.md
- docs/core/TRACKING.md
- bom/bom_estimated_items.csv

Historical provenance

- [INSTALL MINUS 12 READINESS PLAN](#) preserves the May 7 install-window AC planning context; this file owns current Phase 1 AC procurement/implementation.

Physical shore-inlet install gate (2026-07-19)

- The L5-30 inlet remains purchased but not yet cut into the camper/bed-side interface.
- The electrical module is now hard-mounted. The MultiPlus and combined AC breaker enclosure were removed for the lift and must be remounted into the backer's embedded pronged/spiked T-nuts before the final shore path is terminated.
- From the remounted AC-in breaker endpoint, mock the full 10/3 path to the candidate exterior inlet with required bend radius, cable support, drip/water management, service loop, strain relief, and access to both sides of the cut.
- Any small cable opening through the plywood electrical backer requires a correctly sized grommet/bushing or gland plus independent cable support/strain relief; bare 10/3 must not bear on a raw plywood edge.
- Confirm the exterior cover swing, connector clearance, wall/backing thickness, hidden structure/no-drill zone, butyl bedding land, finish-seal geometry, and a path that does not share the wet-side chase.
- Cut only after the complete inside and outside route is proven. Do not let the easiest exterior location create an inaccessible cable entry or service-obscuring bend behind the electrical module.

Locked AC Architecture

AC-in chain (shore to inverter)

- shore source/adapters -> portable 30A EMS -> optional 25 ft TT-30 extension -> 30A locking shore cord -> L5-30 shore inlet -> combined 6-way AC DIN enclosure -> 30A UL489 AC-in breaker/disconnect -> MultiPlus AC-in (L/N/PE)
- AC-in conductors are 10 AWG / 10/3 on the protected AC-in path (30A hardware basis).
- MultiPlus input current limit is set to actual source when adapters are used. Use 10A for first household tests and 12A maximum policy on a normal 15A outlet; do not leave the limit at 50A on adapter/household shore.
- Use the modular extension only when the extra reach is needed. Fully seat, elevate, and weather-protect the TT-30 midpoint connection, and uncoil both cords under load.

AC-out chain (inverter-backed branch distribution)

- MultiPlus AC-out-1 -> 10/3 feeder -> combined 6-way AC DIN enclosure -> 30A UL489 AC-out main breaker -> 20A Branch A + 20A Branch B -> GFCI receptacle per branch
- The two first-in-chain receptacles are 20A self-test GFCIs: Branch A = office/driver side, Branch B = Galley/passenger side. Owner report 2026-08-27 : each GFCI now feeds one downstream duplex from its LOAD side, for four duplex devices/eight plug positions total. Electrical acceptance remains open.
- Inspect all four devices in listed accessible boxes with covers and compatible cable clamps; confirm conductor fill, wall-stack fit, 12 AWG , LINE/LOAD/PE , PE continuity, labels, GFCI trip/reset, whole-chain downstream protection, and representative loads in inverter and accepted-shore modes.
- AC-out branch hardware is not required to perform the initial AC-in-only battery charging test, but it is now included in the purchased Phase 1 cart.

Neutral and ground handling

- AC-in and AC-out neutral termination paths remain isolated.

- Separate AC-in neutral pass-through/termination and AC-out neutral bar are required inside the combined enclosure.
- Common equipment grounding/PE bus is acceptable; do not use it as a neutral.
- Continuous equipment ground path and chassis bond are required end-to-end. The installed MultiPlus-II 48/3000/35-50 120V is a Class I device; its official manual identifies the external M6 connection as the primary PE point, requires at least 4 mm² grounding conductor, and explicitly requires the casing to be connected to the vehicle chassis in a mobile shore-power installation. Use 10 AWG green stranded copper from the MultiPlus external M6 PE lug to a verified truck-chassis bond point; do not leave the lug open and do not use the aluminum Hiatus shell or 80/20 as the only protective-earth path.
- Bond the aluminum camper shell separately into the same chassis/equipment-ground network with a corrosion-compatible connection. Treat shell hardware/body contact as unproven until low-resistance continuity is measured; use compatible/tinned terminals and antioxidant/isolation practice appropriate to the aluminum interface.
- Do not jumper the MultiPlus case/PE lug to Lynx negative or the 12V negative bus. The 48V and 12V DC return paths remain on their dedicated conductors/buses; Mechman case-ground behavior is a separate commissioning check that may establish the single deliberate house-negative-to-chassis reference through the alternator and its dedicated 2/0 return.
- Do not add an always-bonded downstream neutral-ground bond in branch receptacle wiring. Leave the MultiPlus internal ground relay enabled for the normal single-unit topology: it bonds AC-out neutral to chassis in inverter mode and opens that bond when shore AC is accepted.

AC-out-2 policy

- AC-out-2 is **reserve-only** in Phase 1.
- Keep labeled panel space and capped route only; no energized branch hardware is procured for this path in Phase 1.

Required Purchasable Components (Phase 1)

Immediate AC-in-only initial charge path

These rows unblock safe MultiPlus shore charging and should not wait on final receptacle count:

Component class	Qty	Rating/listing requirement	BOM row(s)	Phase 1 status
Shore inlet	1	30A 125V L5-30 shore inlet with exterior cover	107	Purchased
Shore cord, modular extension + household adapter	1 locking cord + 1 extension + 1 dogbone	30A TT-30P-to-L5-30R cord, 25 ft TT-30P-to-TT-30R extension, and 15A household dogbone adapter	108, 340, 281	Purchased
Portable EMS/surge protector	1	Portable 30A EMS with open-neutral/polarity/voltage protection	123	Purchased
AC-in breaker/disconnect	1 of 2	DIN-mount UL 489 1-pole 30A 120VAC	13	Purchased
Combined AC DIN enclosure and bus hardware	1 enclosure + accessories	6-way DIN enclosure; physically isolate AC-in and AC-out neutrals	109, 14	Purchased
Shore + AC-in feed cable	as routed	10/3 stranded tinned-copper cable for C-28/C-29 (30A path)	114	Purchased
Shore-inlet seal/bedding consumables	per install	Butyl bedding + compatible exterior polyurethane perimeter/finish seal	179, 180	Purchased
Strain relief/grommets/clamps/labels/ferrules	per entries	Sized for selected cable/enclosure terminals	38, 41, 43, 44, 45	Cable glands on hand; ferrule kit purchased

Full AC-out distribution / final Phase 1 branch hardware

Component class	Qty	Rating/listing requirement	BOM row(s)	Phase 1 status
Shore inlet	1	30A 125V L5-30 shore inlet with exterior cover	107	Purchased
Shore cord, modular extension + household	1 locking cord + 1 extension + 1	30A TT-30P-to-L5-30R cord, 25 ft TT-30P-to-TT-30R extension, and 15A household dogbone adapter	108, 340, 281	Purchased

adapter	dogbone			
Portable EMS/surge protector	1	Portable 30A EMS used source-side before the shore cord	123	Purchased
AC-in breaker/disconnect	1	DIN-mount UL 489 1-pole 30A 120VAC	13	Purchased
AC-out main breaker	1	DIN-mount UL 489 1-pole 30A 120VAC, ahead of branch breakers	327	Purchased
Combined DIN enclosure	1	6-way DIN enclosure with slot plan: 30A in, 30A out, 20A, 20A, 2x blank/spare	109	Purchased
AC-out branch breaker set	2 active	DIN-mount UL 489 branch breakers: 20A + 20A	110, 330	Purchased
DIN accessory kit	1 kit	Neutral isolation hardware, PE/ground bus support, ferrules, labels/blanks as needed	14, 41	Purchased / labels-blanks confirm on hand
MultiPlus/chassis and shell bonding materials	field-measured	10 AWG green stranded copper minimum for MultiPlus M6 PE to verified truck chassis; separate shell bond jumper; M6/tinned lugs, protected chassis attachment, compatible aluminum-interface hardware/compound	51, 339	Inventory/buy
GFCI receptacles	2	20A self-test GFCI receptacles, one per active branch	15	Purchased
Standard downstream receptacles	2 installed / owner-reported	One downstream duplex from each first GFCI LOAD, same 12 AWG branch, accessible listed box	inactive historical 111; installed inventory source not yet reconciled	Inspect/test as-built
Outlet boxes + covers/faceplates + clamps	4 installed device sets / owner-reported	Six 14 cu in old-work boxes were purchased; prove each used device's listed box, compatible cover/plate and clamp, conductor fill, wall-stack fit, and service access	active row 112 plus inventory	Physical acceptance open
AC branch cable	30 ft purchased	12/3 stranded triplex branch cable (C-31/C-32)	113	Purchased
Shore + AC-in/AC-out feeder cable	20 ft purchased	10/3 stranded triplex for C-28/C-29/C-30 (30A paths)	114	Purchased
Strain relief/cable glands	per enclosure entries	Use assorted on-hand entry hardware sized during physical layout	44	On-hand fitment stock
Grommets	per pass-through points	Abrasion protection at penetrations, selected hands-on during routing	43	On-hand / install fit
P-clamps and retention hardware	per route	Cable support and vibration control, selected hands-on during routing	45	On-hand / install fit
Loom/sleeving	per exposed runs	Harness abrasion protection	42	Required
Heat shrink (adhesive)	install consumable	Termination sealing and strain relief support	38	Required
Ferrules/terminals (AC-relevant)	install consumable	Sized to 10 AWG and 12 AWG terminations as required by device terminals	41, 116	Required

AC-out-2 branch breaker/protection hardware	ø in Phase 1	Reserve-only route, no energized branch hardware in this phase	N/A	Reserve-only (not procured)
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First-Live AC-In Test Result (2026-05-27)

- AC Input 1 should be labeled Shore power for this mobile/source-current-limited system.
- MultiPlus switch II is charger-only; switch I is inverter/charger normal mode; 0 is off.
- Short shore test passed with household-source current limiting: about 1294W shore input and about 54.3V x 21.6A (~1173W) battery charge in bulk.
- Current disconnect practice: connect RV/EMS/load side first, then energize shore; disconnect in reverse by de-energizing/unplugging shore source before disconnecting the RV side.
- This result does not close battery-charge-profile commissioning. Program/verify the MultiPlus lithium settings before sustained charging.

Manual AC Validation Checklist

0) AC-in-only initial charger validation

- Confirm AC-in physical order: shore source/adapters -> portable EMS -> optional TT-30 extension -> locking shore cord -> inlet -> AC-in breaker/disconnect -> MultiPlus AC-in .
- Confirm AC-out breakers/loads are disconnected or not yet installed for the first battery-charge test.
- Confirm MultiPlus input current limit is set to actual source (10A first household test, 12A max on normal 15A , actual rating for 20A / 30A).
- Confirm AC Input 1 is labeled/configured as Shore power .
- Confirm battery charge profile is intentionally set for the selected LiFePO4 voltage basis before sustained charging.
- Confirm AC-in and AC-out neutral paths are not mixed.

Use this checklist as the continuing acceptance gate before sustained shore charging, AC-out branch energization, or any AC enclosure closeout.

1) Topology integrity

- Confirm one unique AC-in chain exists: shore source/adapters -> portable EMS -> cord -> inlet -> AC-in breaker -> MultiPlus AC-in .
- Confirm one unique AC-out-1 chain exists: MultiPlus AC-out-1 -> 30A AC-out main -> 20A branch breakers -> GFCI receptacles .
- Confirm AC-out-2 is documented as reserve-only and not active in Phase 1 procurement.
- Confirm the two first-in-chain GFCIs remain the only branch-origin devices and each owner-reported downstream duplex is fed only from its corresponding GFCI LOAD ; prove compatible listed boxes/covers/clamps, conductor fill/accessibility, and whole-chain testing.

2) Protection coordination

- Confirm AC-in breaker is 30A and AC-in conductors are 10 AWG .
- Confirm AC-out main breaker is 30A and the MultiPlus-to-enclosure feeder is 10 AWG .
- Confirm branch OCP values are 20A and 20A with 12 AWG branch conductors.
- Confirm breaker listing basis is UL 489 (or equivalent NRTL listing) for branch/feeder use.

3) Neutral/ground correctness

- Confirm AC-in and AC-out neutral paths are isolated.
- Confirm input and output neutral paths are separately landed/passed through inside the combined enclosure.
- Confirm the equipment grounding/PE path is common and continuous.
- Confirm the MultiPlus external M6 PE lug is bonded to a verified truck-chassis point with at least 4 mm² / selected 10 AWG green stranded copper; the aluminum shell/80/20 is not the sole return.
- Confirm the aluminum shell has its own corrosion-compatible bond to the chassis/equipment-ground network and verify low-resistance continuity after final assembly.
- Confirm no intentional jumper exists from MultiPlus case/PE to Lynx negative or the 12V negative bus, and confirm no chassis path bypasses the SmartShunt after the Mechman ground style is physically identified.
- Confirm no fixed downstream neutral-ground bond is added in branch receptacle wiring.
- Confirm the MultiPlus internal ground relay remains enabled for this normal single-unit mobile topology; test both GFCI branches in inverter mode and on accepted shore power before routine use.

4) Procurement completeness

- Confirm every required AC component class has a BOM row mapping.
- Confirm no AC-critical component exists only as implied text.

- Confirm AC-out-2 hardware remains excluded from Phase 1 carts.

5) Documentation parity

- Confirm AC assumptions match across:
 - docs/implementation/ELECTRICAL_AC_BOM.md
 - docs/implementation/ELECTRICAL_overview_diagram.md
 - docs/core/SYSTEMS.md
 - docs/core/TRACKING.md
 - bom/bom_estimated_items.csv

6) Operating scenarios

- AC-in-only initial charge: no AC-out loads connected, MultiPlus charger behavior documented.
- Shore present (30A source): pass-through + charging behavior documented.
- Shore present via 15A/20A adapter: current-limit setting policy documented.
- Shore absent: inverter-backed AC-out-1 behavior documented.
- GFCI trip/reset behavior per branch is called out for commissioning test.

Procurement Notes

- DIN rail is a mounting method; breaker listing and rating remain the controlling requirement.
- Lowest-cost listed policy is acceptable only if each selected device has verifiable NRTL listing (UL or ETL) for intended use.
- Purchased SKU lock is recorded in bom/bom_estimated_items.csv for rows 13 , 327 , 14 , 15 , 41 , 107 , 108 , 109 , 110 , 112 , 330 , 113 , 114 , 123 , 179 , and 180 . Rows 13 / 327 are the AC-in/AC-out 30A pair; rows 110 / 330 are the two 20A branch breakers. Row 112 is the purchased old-work box six-pack, not evidence that six downstream devices are approved. Row 181 was same-order tire-deflator/off-road support hardware and is intentionally outside AC scope.
- Inactive BOM row 111 preserves the formerly closed downstream-receptacle concept. Row 112 supplied the six purchased old-work boxes. Owner report 2026-08-27 closes physical device count at two GFCIs plus two downstream duplexes; installed receptacle/cover/clamp inventory provenance and all box-fill/wall-stack/accessibility/testing gates remain open.
- Current utilization note (2026-05-16): AC protection chain and purchase scope are locked for a 30A system with two active 20A branches. Do not add a third active AC branch without revisiting the AC-out main/enclosure/feed plan.